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Predicting module-lattice reduction

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Motivation

Lattice problems underpin most of post-quantum cryptography

- NIST standards: ML-KEM (Kyber), ML-DSA (Dilithium), FN-DSA (Falcon)
- Based on **module lattices**: Equipped with algebraic **structure** inherited from their underlying **number field**

Central cryptanalytic tool: **Lattice reduction**, such as the **BKZ algorithm** [Sch87, SE94]

- *General belief*: BKZ does not “see” the structure

What if we modify BKZ to **exploit** the structure? *Does such a **module-BKZ** algorithm give an advantage?*

- First posed as “Q8” in Kyber’s submission to NIST
- If answer is “yes”, it could affect concrete security estimates (also beyond Kyber!)

Core question

in practice

*Does module-BKZ perform **better** or **worse** than unstructured BKZ? By how much?*

Prior work:

- [LPSW19, MS19]: Defined **module-lattice versions** of lattice reduction and studied their **worst-case** behavior
 - Purely worst-case asymptotic guarantees, which can be quite different from practical performance!
- [KK24]: First **experimental** study of module-LLL on NTRU lattices
 - Results limited to LLL and no predictive framework

Short answer

*It depends on the number field
Key driver: Discriminant*

Our contribution:

- Systematic **heuristic-based** study of module-BKZ's **average-case** performance
- Supported by **experiments**

Outline of the talk

- I. **Background** — Lattices, BKZ reduction, and module lattices
- II. **Module-BKZ** — Definition and profile prediction
- III. **Analysis** — Theory and experiments on the module-BKZ profile
- IV. **Impact** — Consequences for cryptanalysis and open questions
- V. **Conclusion** — Main takeaways

Background

Lattices, BKZ reduction, and module lattices

Lattices and lattice reduction

(Euclidean) lattice in $\mathbb{R}^n$ of dimension k : A set of the form $L = \left\{ \sum_{i=1}^k z_i \mathbf{b}_i \mid z_1, \dots, z_k \in \mathbb{Z} \right\}$ for linearly independent $\mathbf{b}_1, \dots, \mathbf{b}_k \in \mathbb{R}^n$

- $\mathbf{b}_1, \dots, \mathbf{b}_k$ is a **basis** (or $\mathbb{Z}$ -basis) of L
- **GSO:** The “Gram-Schmidt” vectors $\mathbf{b}_1^*, \dots, \mathbf{b}_k^*$ defined by $\mathbf{b}_i^* = \pi_i(\mathbf{b}_i)$, where π_i is the projection orthogonal to $\text{span}_{\mathbb{R}}(\mathbf{b}_1, \dots, \mathbf{b}_{i-1})$
- **Determinant:** $\det_{\mathbb{Q}}(L) = \prod_{i=1}^n \|\mathbf{b}_i^*\|$

Shortest Vector Problem (SVP): Given a basis of L , find nonzero $\mathbf{v} \in L$ with minimal Euclidean norm $\lambda_1(L)$

- Difficulty of solving SVP (and other lattice problems) depends on which basis you are given
 - If the basis is close to **orthogonal**, it will “reveal” short vectors; otherwise it may “hide” them

Quality of a basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ is measured by its $\mathbb{Q}$ -**profile** $(\ell_1^{\mathbb{Q}}, \dots, \ell_k^{\mathbb{Q}})$, where $\ell_i^{\mathbb{Q}} := \ln \|\mathbf{b}_i^*\|$

- Note: $\sum_{i=1}^n \ell_i^{\mathbb{Q}} = \ln \det_{\mathbb{Q}}(L)$
- **More balanced = better**

*Turning a lattice basis into one of better quality is the task of **lattice reduction***

BKZ algorithm

Main idea: Improve the basis by repeatedly solving SVP in **small blocks** of basis vectors

Input: Basis of an n -dimensional lattice L
 Parameter $\beta \geq 2$ (“blocksize”)

Output: **BKZ $^\beta$ -reduced** basis $\mathbf{B}$, i.e., for all $1 \leq i \leq n$, $\|\mathbf{b}_i^*\| = \lambda_1(L(\mathbf{B}_{[i, \min(i+\beta-1, n)]}))$

Skeleton:

↖ *Projected lattice generated by*
 $\pi_i(\mathbf{b}_i), \dots, \pi_i(\mathbf{b}_{\min(i+\beta-1, n)})$

1. **Pick a block** of (at most) β consecutive basis vectors
2. **Project** this block onto the orthogonal complement of earlier basis vectors
3. **Solve SVP** in the corresponding projected lattice
4. **Insert** short vector back into the basis (and remove linear dependencies)
5. **Repeat** with a new block until succeeded

← *Using an SVP oracle in dimension β*

Tradeoff: Larger β means flatter $\mathbb{Q}$ -profile, but also higher runtime

BKZ profile prediction

What can we say about the **quality** of a BKZ^β -reduced basis?

↖ *For random lattices*

Observation (for large β): After BKZ, $\mathbb{Q}$ -profiles look “geometric”

Geometric Series Assumption (GSA) as formalized in [Sch03]:

Let $\beta \ll n$ be sufficiently large. There is a constant $\alpha > 1$ such that the $\mathbb{Q}$ -profile of a BKZ^β -reduced basis of a **random n -dimensional lattice** of fixed determinant satisfies

$$\mathbb{E}[\ell_i^{\mathbb{Q}}] = \mathbb{E}[\ell_1^{\mathbb{Q}}] - (i - 1)\ln \alpha \quad \forall 1 \leq i \leq n.$$

$\implies$ Slope of $\mathbb{Q}$ -profile is $-\ln \alpha$

Slope prediction: The $\mathbb{Q}$ -profile of a BKZ^β -reduced basis $\mathbf{B}$ of a random lattice L satisfies

$$\text{slope}_{\mathbb{Q}}(\text{BKZ}^\beta) = -\frac{2}{\beta - 1} \mathbb{E}_{\mathbf{s}}[\ln \|\mathbf{s}\|]$$

where $\mathbf{s}$ is a shortest vector in (the normalization of) one of the projected lattices $L(\mathbf{B}_{[i, i+\beta-1]})$

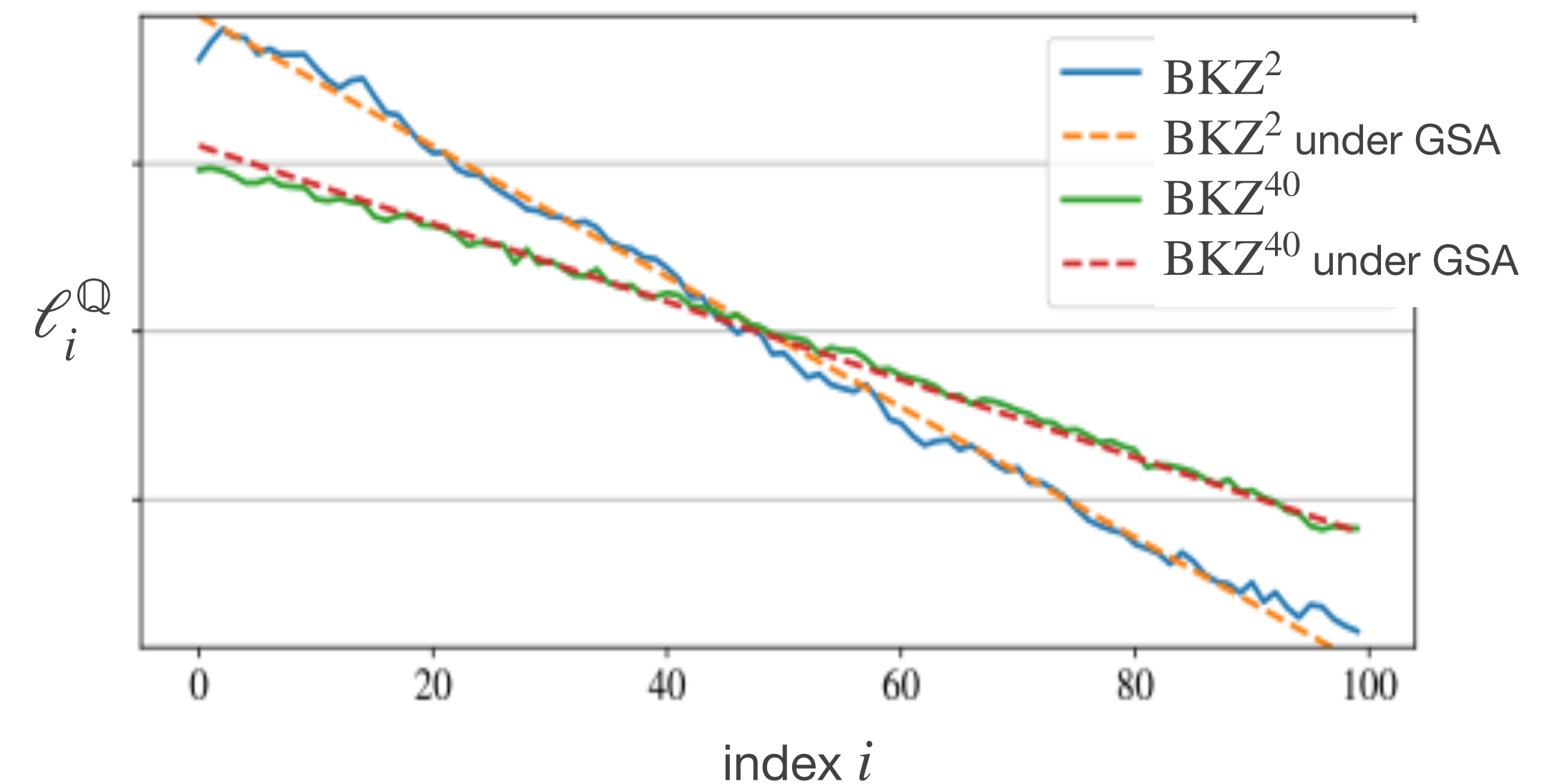


Figure 2.8 in ePrint 2025/304
(de Boer and van Woerden)

BKZ profile prediction under GSA + GH

What is the expected (log-)norm of a shortest vector in a **random** lattice?

Gaussian Heuristic (GH) based on [Söd11, Che13]:

*Let L be a random β -dimensional lattice of unit determinant.
Then the Gaussian Heuristic predicts*

$$\mathbb{E}[\ln \lambda_1(L)] = \text{lgh}_{\mathbb{Q}}(\beta) := \frac{1}{\beta} \left(\ln \frac{2}{\text{vol}(\mathcal{B}_{\beta})} - \gamma \right)$$

where γ denotes the Euler-Mascheroni constant and $\mathcal{B}_{\beta}$ the β -dimensional Euclidean unit ball.

Together, **GSA + GH** form a widely used **predictive model** of BKZ performance (for random lattices):

The $\mathbb{Q}$ -profile of a BKZ^{β} -reduced basis satisfies

$$\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta}) = -\frac{2}{\beta-1} \text{lgh}_{\mathbb{Q}}(\beta)$$

Module lattices

Euclidean lattices “live” in Euclidean space

- While module lattices can be viewed as Euclidean lattices, they have a **richer algebraic structure**
- Depends on their underlying **number field**

Relevant notation:

- **Number field** is a field extension $K \supseteq \mathbb{Q}$ of finite **degree** d
- $\mathcal{O}_K$ denotes the **ring of integers** of K
- Δ_K denotes the **discriminant** of K
- We will mainly work in the ring $K_{\mathbb{R}} = K \otimes_{\mathbb{Q}} \mathbb{R}$
- Two notions of “length” for vectors in $K_{\mathbb{R}}^r$:
 - Geometric **trace norm** $\|\cdot\|$
 - **Algebraic norm** $N(\cdot)$, which can also be extended to fractional ideals $I \subseteq K$

*Example: **Cyclotomic field** of **conductor** c*

- $K = \mathbb{Q}(\omega_c)$ where ω_c is a primitive c -th root of unity
- Its degree is $d = \phi(c)$
- Ring of integers is $\mathcal{O}_K = \mathbb{Z}[\omega_c]$

Module lattices

A **rank- r module** over a number field K of degree d :

- $M = \sum_{i=1}^r \mathbf{b}_i I_i$ for $K_{\mathbb{R}}$ -linearly independent $\mathbf{b}_1, \dots, \mathbf{b}_r \in K_{\mathbb{R}}^r$ and nonzero fractional ideals $I_1, \dots, I_r$
- We call $\mathfrak{B} = (\mathbf{b}_i, I_i)_{i=1}^r$ a **pseudobasis** of M
- Gram-Schmidt orthogonalization can be extended to $K_{\mathbb{R}}^r$, we write $\mathbf{b}_1^*, \dots, \mathbf{b}_r^*$ for the **GSO**

There are d distinct injective field homomorphisms $\sigma: K \rightarrow \mathbb{C}$, called the **embeddings** of K

Through these embeddings we can view M as an rd -dimensional Euclidean lattice that **preserves the geometry**:

- | | | | |
|-------------------|--------------------|---------------|--|
| ▸ Space: | $K_{\mathbb{R}}^r$ | $\rightarrow$ | $\mathbb{R}^{rd}$ |
| ▸ Rank: | module rank r | $\rightarrow$ | lattice dimension rd |
| ▸ Representation: | pseudobasis | $\rightarrow$ | $\mathbb{Z}$ -basis in $\mathbb{R}^{rd}$ |
| ▸ Geometry: | trace norm | $\rightarrow$ | Euclidean norm |

Determinant of the module lattice M is $\det_{\mathbb{Q}}(M) = |\Delta_K|^{r/2} \prod_{i=1}^r N(\mathbf{b}_i^*) N(I_i)$

Module-BKZ

Definition and profile prediction

Module-BKZ

BKZ treats a **rank- r module M** just as an **rd -dimensional Euclidean lattice**

$$d = \deg(K)$$

- “Block” = Integer span of β basis vectors (after projection)

Module-BKZ is obtained by taking the **algebraic structure** into account

- Blocksizes are restricted to multiples of the degree: $\beta = \beta_K d$
- “Block” = Span of a rank- β_K **module** (which is a β -dimensional Euclidean lattice)

Same skeleton as BKZ:

Pick a block of consecutive basis vectors $\rightarrow$ Call SVP oracle on projected lattice $\rightarrow$ Insert short vector $\rightarrow$ Repeat

Main difference: Blocks now **respect the algebraic structure of K**

Algorithm and implementation

$\mathfrak{B}$ is **mBKZ $_{\beta_K}^{\beta_K}$ -reduced** if, for all $1 \leq i \leq r$,

$$\|\mathbf{b}_i^*\| = \lambda_1(M(\mathfrak{B}_{[i, \min(i+\beta_K-1, r)]}))$$

where $M(\mathfrak{B}_{[i, \min(i+\beta_K-1, r)]})$ is a natural module analog of $L(\mathbf{B}_{[i, \min(i+\beta-1, n)]})$

Algorithm 1: Module-BKZ (high-level)

Input: Pseudobasis $\mathfrak{B}$ of a rank- r module M ; parameter β_K

Output: mBKZ $_{\beta_K}^{\beta_K}$ -reduced pseudobasis

```

while  $\mathfrak{B}$  is not mBKZ $_{\beta_K}^{\beta_K}$ -reduced do
  for  $i = 1, \dots, r$  do
    Let  $\mathbf{v} = \text{SVP}(M(\mathfrak{B}_{[i, \min(i+\beta_K-1, r)]}))$ 
    Let  $I$  be such that  $\mathbf{v}I = \mathbf{v}K \cap M(\mathfrak{B}_{[i, \min(i+\beta_K-1, r)]})$ 
    Lift  $\mathbf{v}I$  to  $\mathbf{w}I \subseteq M$  satisfying  $\pi_i(\mathbf{w}) = \mathbf{v}$ 
    Insert  $(\mathbf{w}, I)$  into  $\mathfrak{B}$  at position  $i$  and remove linear dependencies
  return  $\mathfrak{B}$ 

```

We provided a Python implementation for several cyclotomic fields K

- Built on top of existing libraries: `fp111`, `G6K` ← Ideally replaced by a proper module-lattice library
- As these are designed for $\mathbb{Z}$ -bases, we perform some tweaks:
 - By “**restructuring**” we ensure the module structure is preserved (or retrieved) at several stages of the algorithm
 - Using the **cyclic embedding** we represent the lattice with integer coefficients while still preserving the right geometry

Profile observations

Let $(\mathbf{b}_i, I_i)_{i=1}^r$ be a pseudobasis of a rank- r module M with GSO $\mathbf{b}_1^*, \dots, \mathbf{b}_r^*$

Let $\mathbf{z}_1^*, \dots, \mathbf{z}_{rd}^*$ be the GSO of a $\mathbb{Z}$ -basis obtained by embedding K into $\mathbb{R}^d$

$\mathbb{Q}$ -profile: $(\ell_1^{\mathbb{Q}}, \dots, \ell_{rd}^{\mathbb{Q}})$ where $\ell_i^{\mathbb{Q}} := \ln \|\mathbf{z}_i^*\|$

- Each entry = log-norm of **one embedded vector**

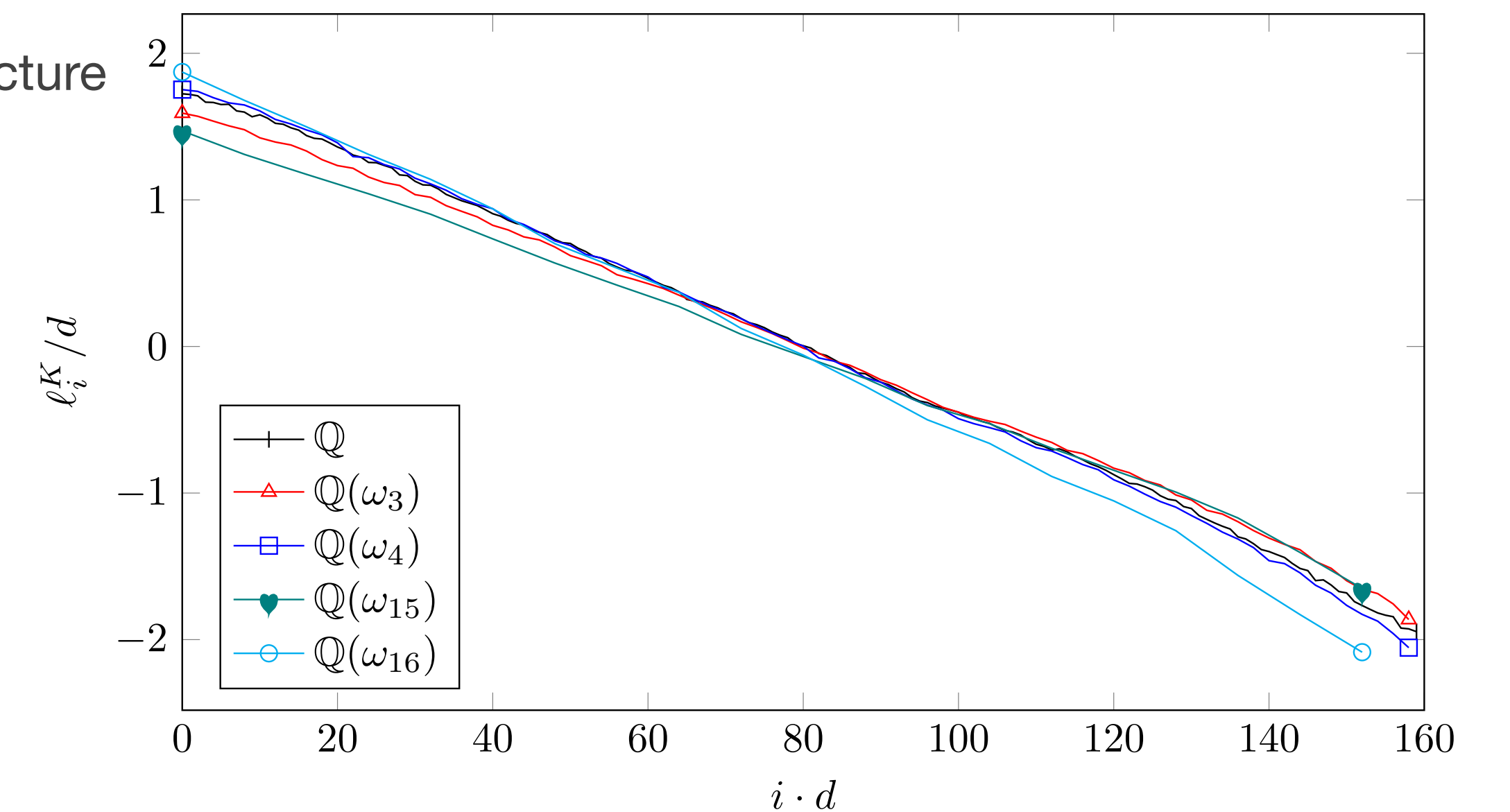
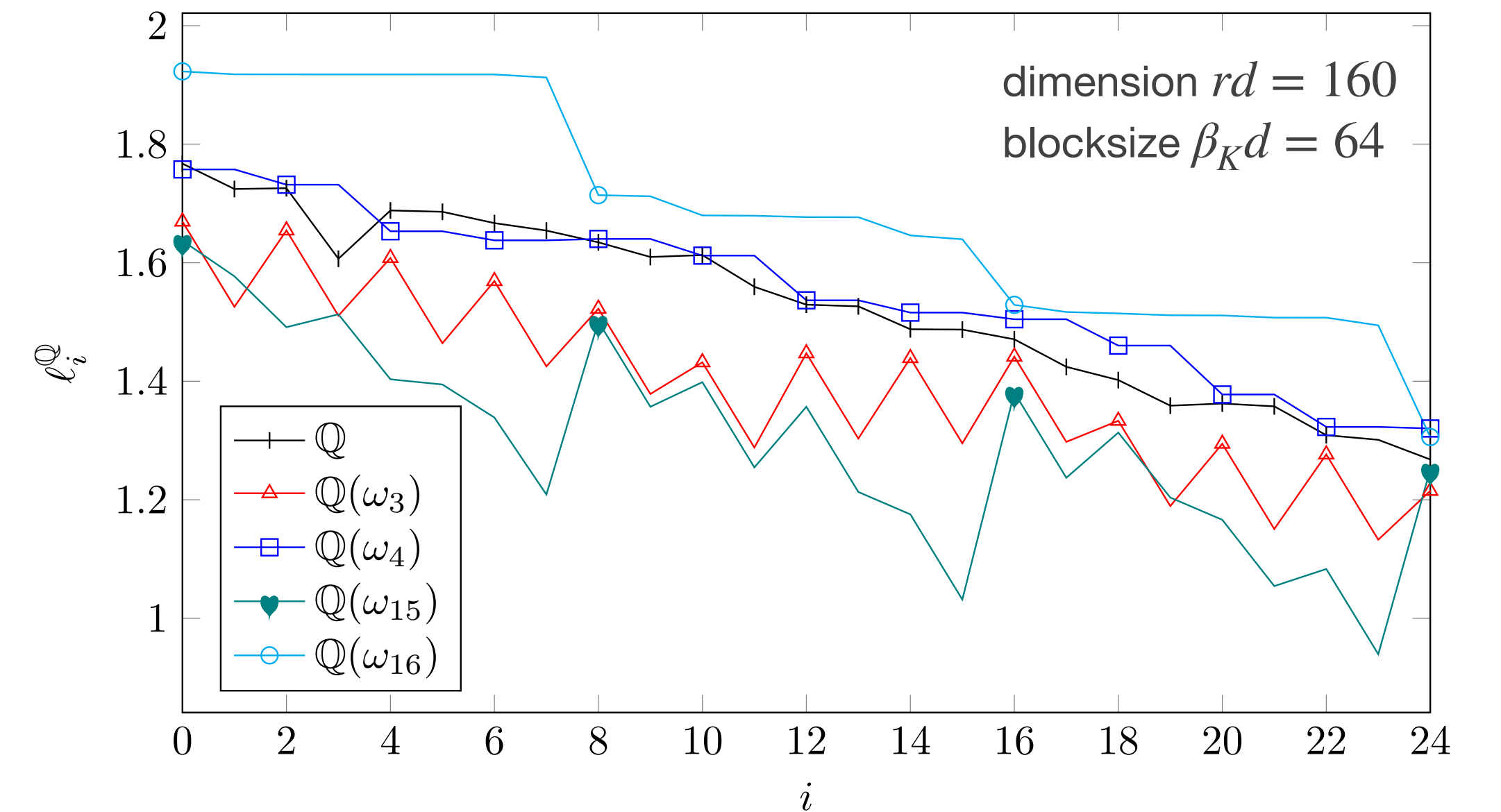
Observation: After module-BKZ, $\mathbb{Q}$ -profiles show **periodic** patterns

- Module-BKZ considers d embedded vectors at once, respecting their structure
- Natural to group these $\mathbb{Q}$ -profile entries together...

K -profile: $(\ell_1^K, \dots, \ell_r^K)$ where $\ell_i^K := \sum_{j=1}^d \ell_{(i-1)d+j}^{\mathbb{Q}} = \ln \det_{\mathbb{Q}}(\mathbf{b}_i^* I_i)$

- Each entry = log-determinant of a **rank-1 module** (d embedded vectors)

Observation: After module-BKZ, K -profiles look “geometric” (for large β_K)



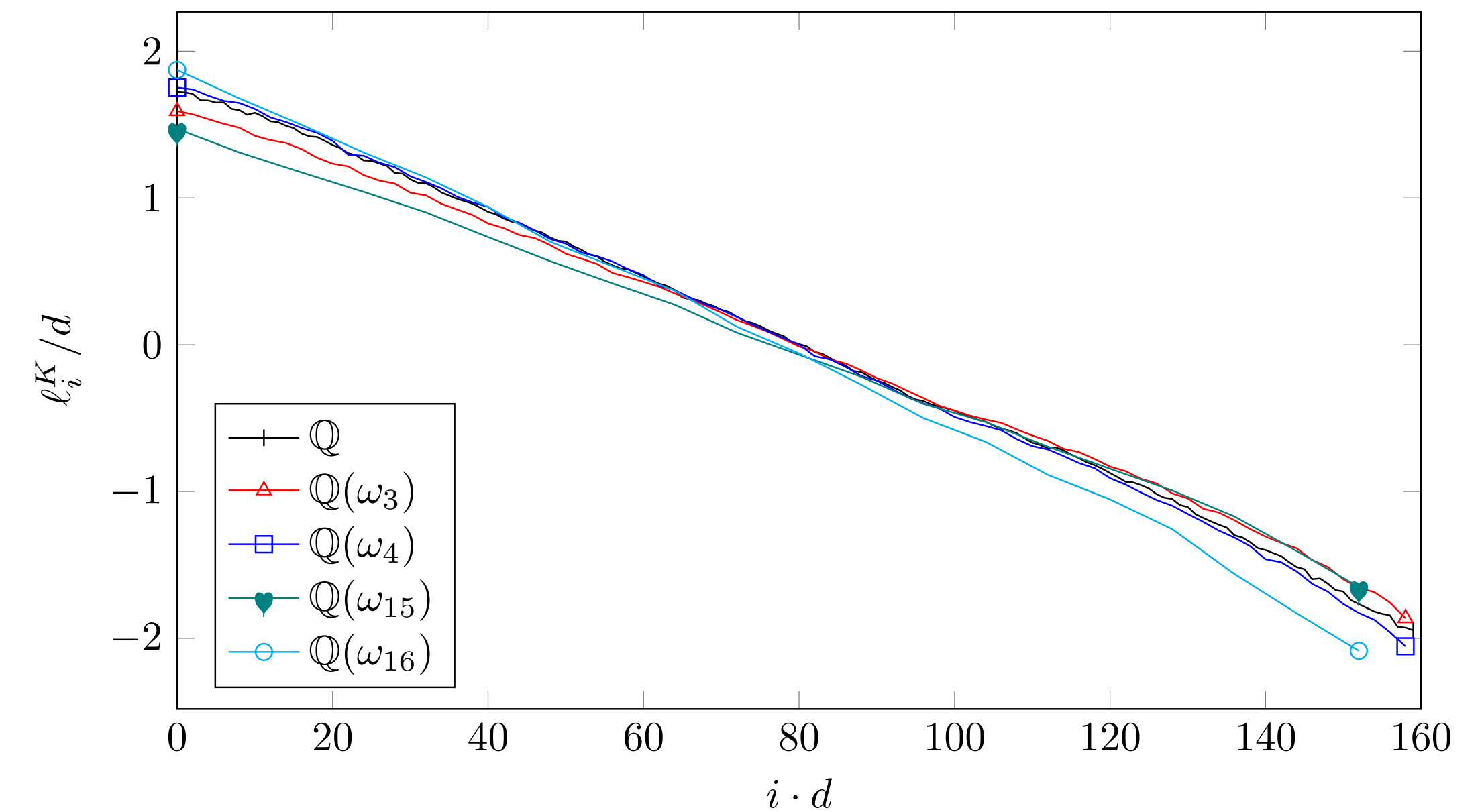
Module-BKZ profile prediction

Recall: $\ell_i^K := \sum_{j=1}^d \ell_{(i-1)d+j}^{\mathbb{Q}} = \ln \det_{\mathbb{Q}}(\mathbf{b}_i^* I_i)$

This motivates introducing the **module-lattice GSA**:

Let K be a number field and $\beta_K \ll r$ sufficiently large. There is a constant $\alpha_K > 1$ such that the K -profile of an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis of a **random rank- r module M over K** of fixed determinant satisfies

$$\mathbb{E}[\ell_i^K] = \mathbb{E}[\ell_1^K] - (i-1) \ln \alpha_K \quad \forall 1 \leq i \leq r.$$



Slope prediction: The $\mathbb{Q}$ -profile of an $\text{mBKZ}_K^{\beta_K}$ -reduced pseudobasis $\mathfrak{B}$ of a random module M satisfies

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = -\frac{2}{\beta_K d - d} \left(\frac{1}{2d} \ln(|\Delta_K|) + \frac{1}{d} \mathbb{E}_{\mathbf{s}}[\ln N(\mathbf{s})] + \frac{1}{d} \mathbb{E}_I[\ln N(I)] \right)$$

where $\mathbf{s}$ is a shortest vector in (the normalization of) one of the projected modules $M(\mathfrak{B}_{[i, i+\beta_K-1]})$ and $I \supseteq \mathcal{O}_K$ an ideal in $\mathfrak{B}$

Module-BKZ profile prediction

For an $\text{mBKZ}_K^{\beta_K}$ -reduced basis:

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = -\frac{2}{\beta_K d - d} (t_1 + t_2 + t_3 + t_4)$$

where:

$t_1 = \mathbb{E}_{\mathbf{s}} [\ln \ \mathbf{s}\]$	module-lattice analog of GH
$t_2 = \frac{1}{2d} \ln \frac{ \Delta_K }{d^d}$	discriminant gap
$t_3 = \mathbb{E}_{\mathbf{s}} \left[\ln \frac{\sqrt{d} N(\mathbf{s})^{1/d}}{\ \mathbf{s}\ } \right]$	skewness gap (≤ 0)
$t_4 = \frac{1}{d} \mathbb{E}_I [\ln N(I)]$	index gap (≤ 0)

For a BKZ^{β} -reduced basis:

$$\text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta}) = -\frac{2}{\beta - 1} \text{lgh}_{\mathbb{Q}}(\beta)$$

*Naturally compared for $\beta := \beta_K d$
(same dimension SVP oracle)*

*larger slope
=
flatter profile
=
better basis*

Analysis

Theory and experiments on the module-BKZ profile

t_1 : Module-Lattice Gaussian Heuristic

What is the expected (log-)norm of a shortest vector in a **random** module over K ?

- For $K = \mathbb{Q}$, we can invoke the Gaussian Heuristic
- [GSVV24]: Concentration results for $\mathbb{E}[\lambda_1(M)]$ for random modules over cyclotomic fields

Module-Lattice Gaussian Heuristic (mGH):

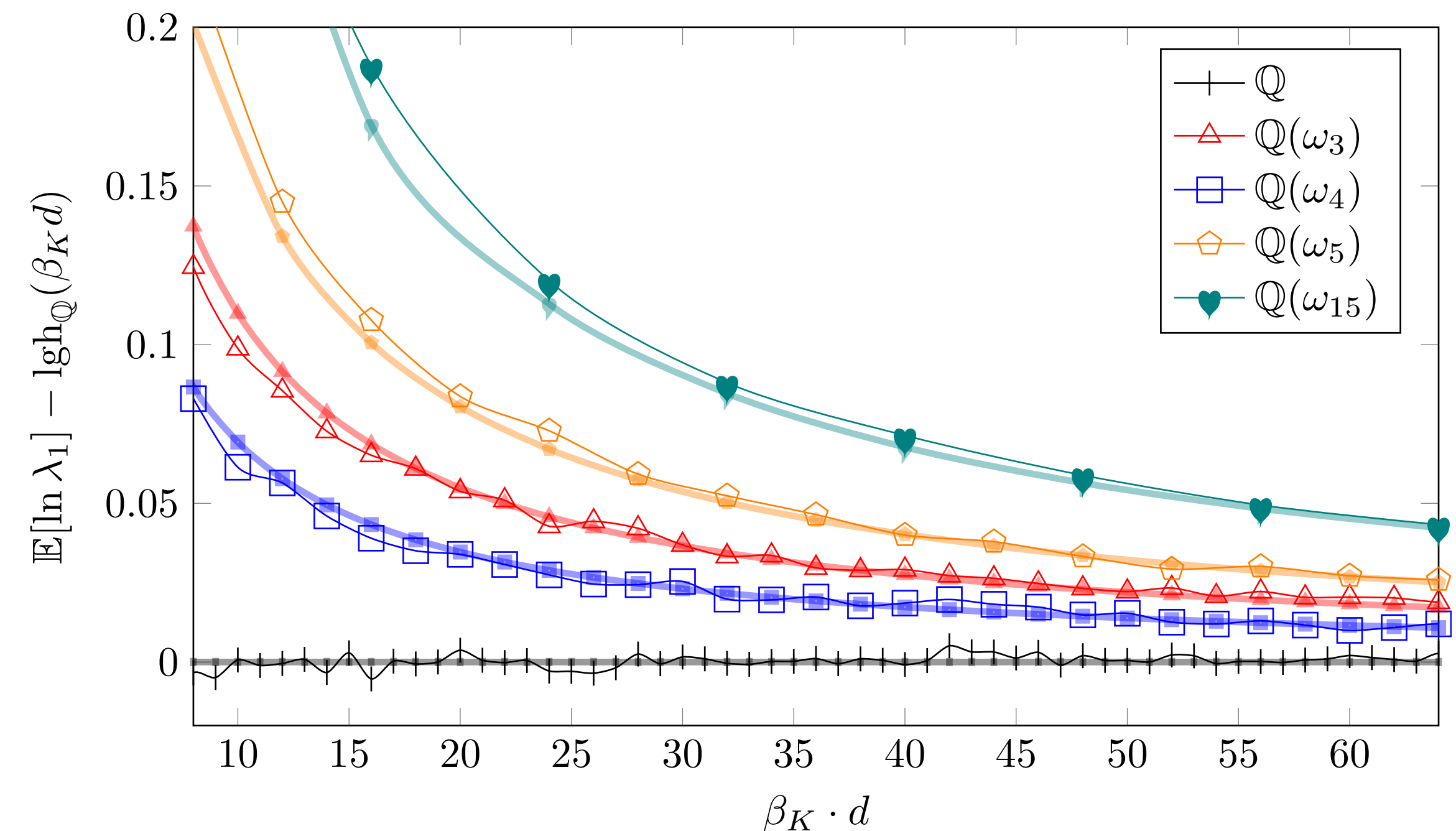
Let M be a random rank- β_K module of unit determinant.
Then the Module-Lattice Gaussian Heuristic predicts:

$$\mathbb{E}[\ln \lambda_1(M)] = \text{lgh}_{\mathbb{Q}}(\beta_K d) + \frac{1}{\beta_K d} \ln \frac{\mu_K}{2}$$

where μ_K denotes the number of roots of unity in K .

We heuristically predict:

$$t_1 = \text{lgh}_{\mathbb{Q}}(\beta_K d) + \frac{1}{\beta_K d} \ln \frac{\mu_K}{2}$$



thick lines: predictions

thin lines: experimental average (over 1000 samples)

t_2 : Discriminant gap

By definition, the discriminant gap depends only on K :

$$t_2 = \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d}$$

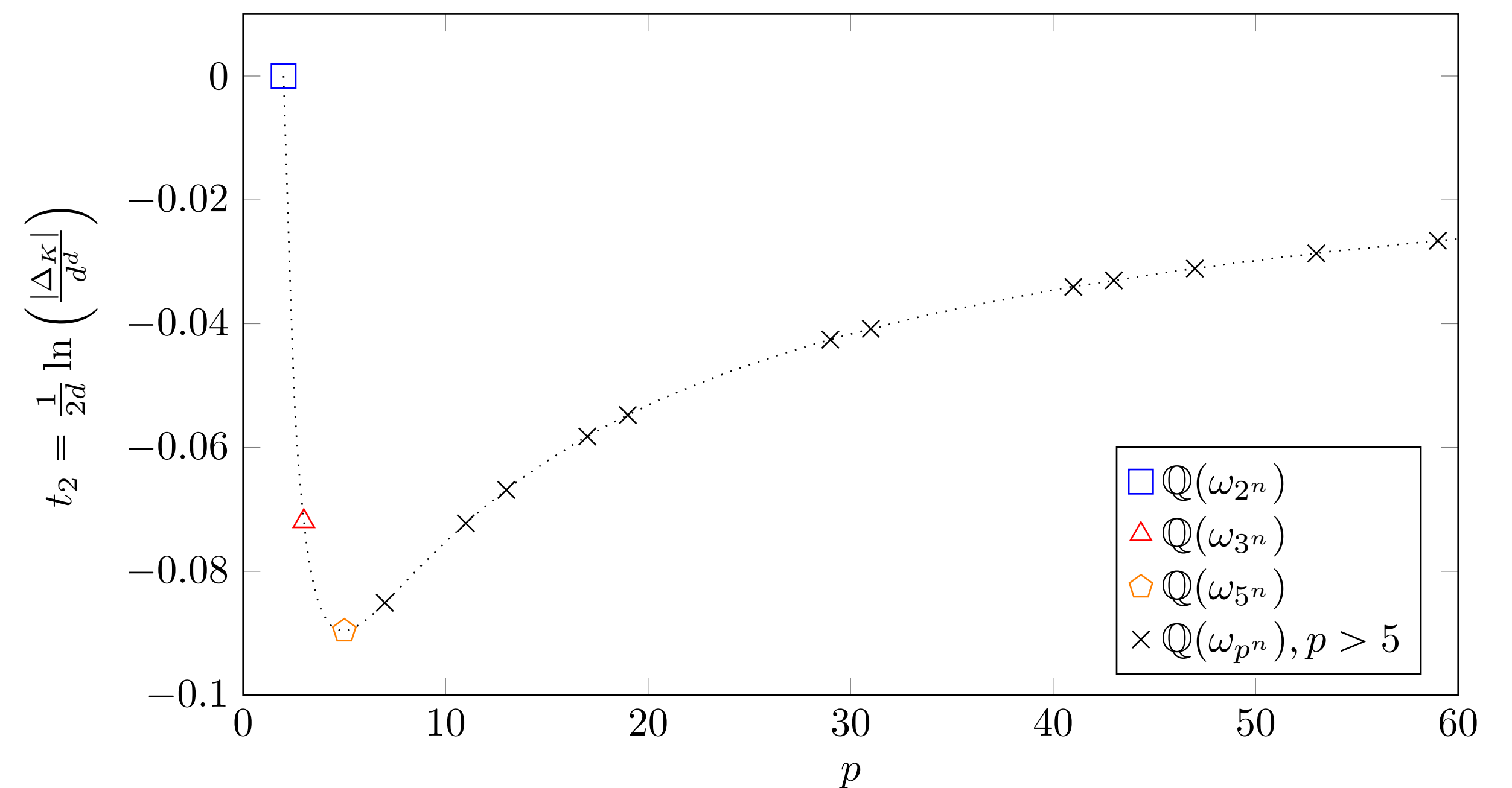
← Constant in β_K

For a **cyclotomic** field $K = \mathbb{Q}(\omega_c)$, we have $t_2 \leq 0$:

- If $c = 2^n$: $|\Delta_K| = d^d$, so $t_2 = 0$
- Otherwise: $|\Delta_K| < d^d$, so $t_2 < 0$

It depends only on $P_c = \{\text{distinct prime factors of } c\}$:

$$t_2 = \frac{1}{2} \sum_{p \in P_c} \left(\frac{p-2}{p-1} \ln(p) - \ln(p-1) \right)$$



Each distinct odd prime factor contributes to **decreasing** $t_2 \implies \text{increasing } \text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) \implies \text{flatter profile}$

- Small odd primes contribute the most

t_3 : Skewness gap

By definition, $t_3 = \mathbb{E}_{\mathbf{s}} \left[\ln \frac{\sqrt{d} N(\mathbf{s})^{1/d}}{\|\mathbf{s}\|} \right]$

where $\mathbf{s} \in K_{\mathbb{R}}^{\beta_K}$ is a shortest vector in a random rank- β_K module

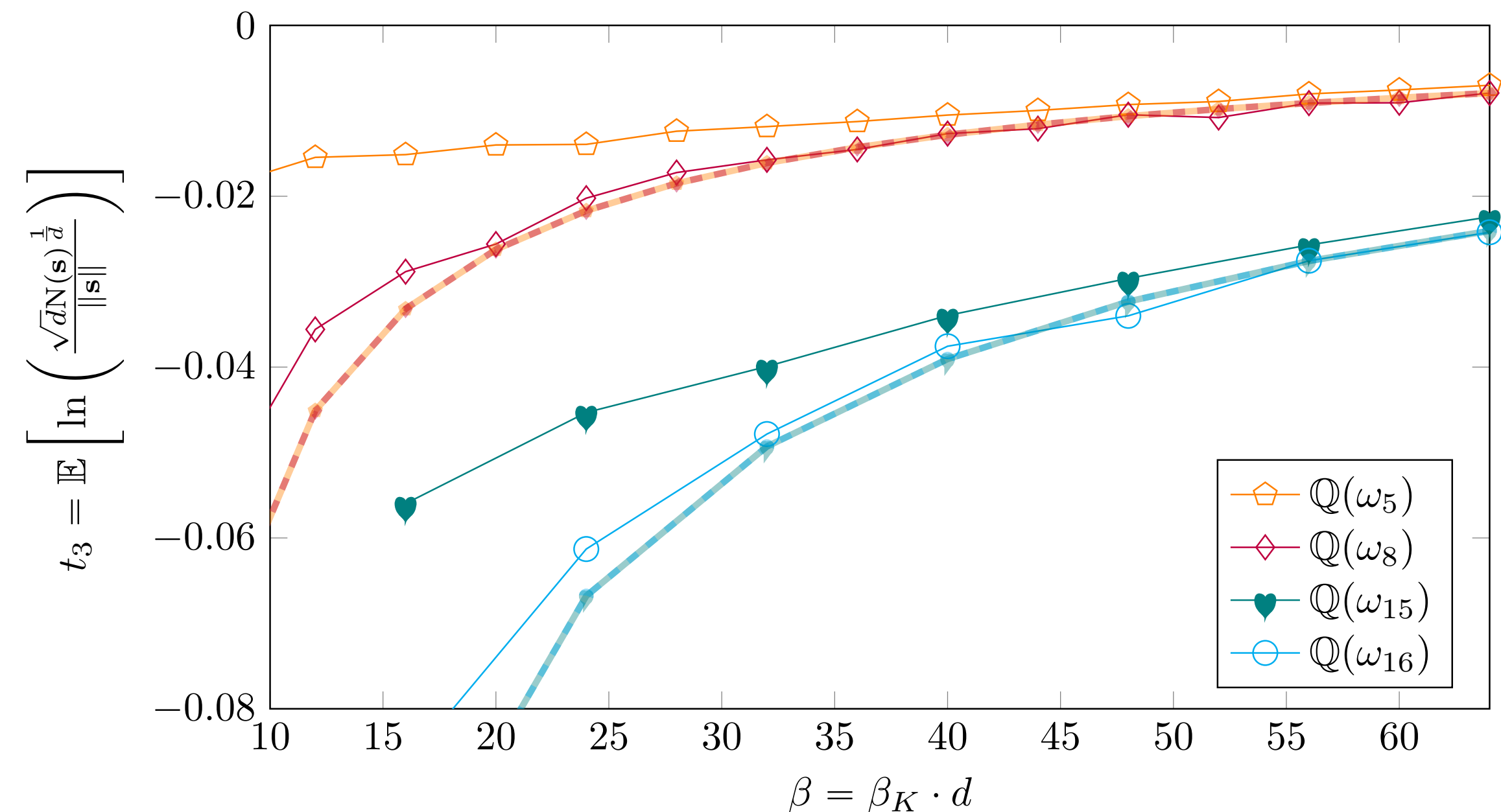
Properties:

- **Arithmetic-geometric inequality** gives $t_3 \leq 0$
- $t_3 = 0$ iff $|\sigma(\mathbf{s})|^2$ is the same for each **embedding** σ of K
 - It measures how “**skewed**” (unbalanced) these values are
- Invariant by scaling $\|\mathbf{s}\|$, so only the direction matters

Spherical model: Assume $\mathbf{s}$ is uniformly random on the unit sphere in $K_{\mathbb{R}}^{\beta_K}$

- **Prediction** $\tilde{t}_3(K, \beta_K)$ in terms of β_K and the number of real and complex embeddings of K
- NB: This forgets that $\mathbf{s}$ is a shortest vector of a **module** over K , which should constrain its direction

Empirical observation: t_3 converges to $\tilde{t}_3(K, \beta_K)$ for large β_K and $\tilde{t}_3(K, \beta_K) \leq t_3 \leq 0$ for small β_K



thick lines: prediction $\tilde{t}_3(K, \beta_K)$

thin lines: experimental average (over 1000 samples)

$$\tilde{t}_3(K, \beta_K) \leq t_3 \leq 0$$

t_4 : Index gap

By definition, $t_4 = \frac{1}{d} \mathbb{E}_I[\ln N(I)]$ where I is one of the fractional ideals in the (random) pseudobasis $\mathfrak{B}$

Algorithm ensures $I \supseteq \mathcal{O}_K$, so $t_4 \leq 0$

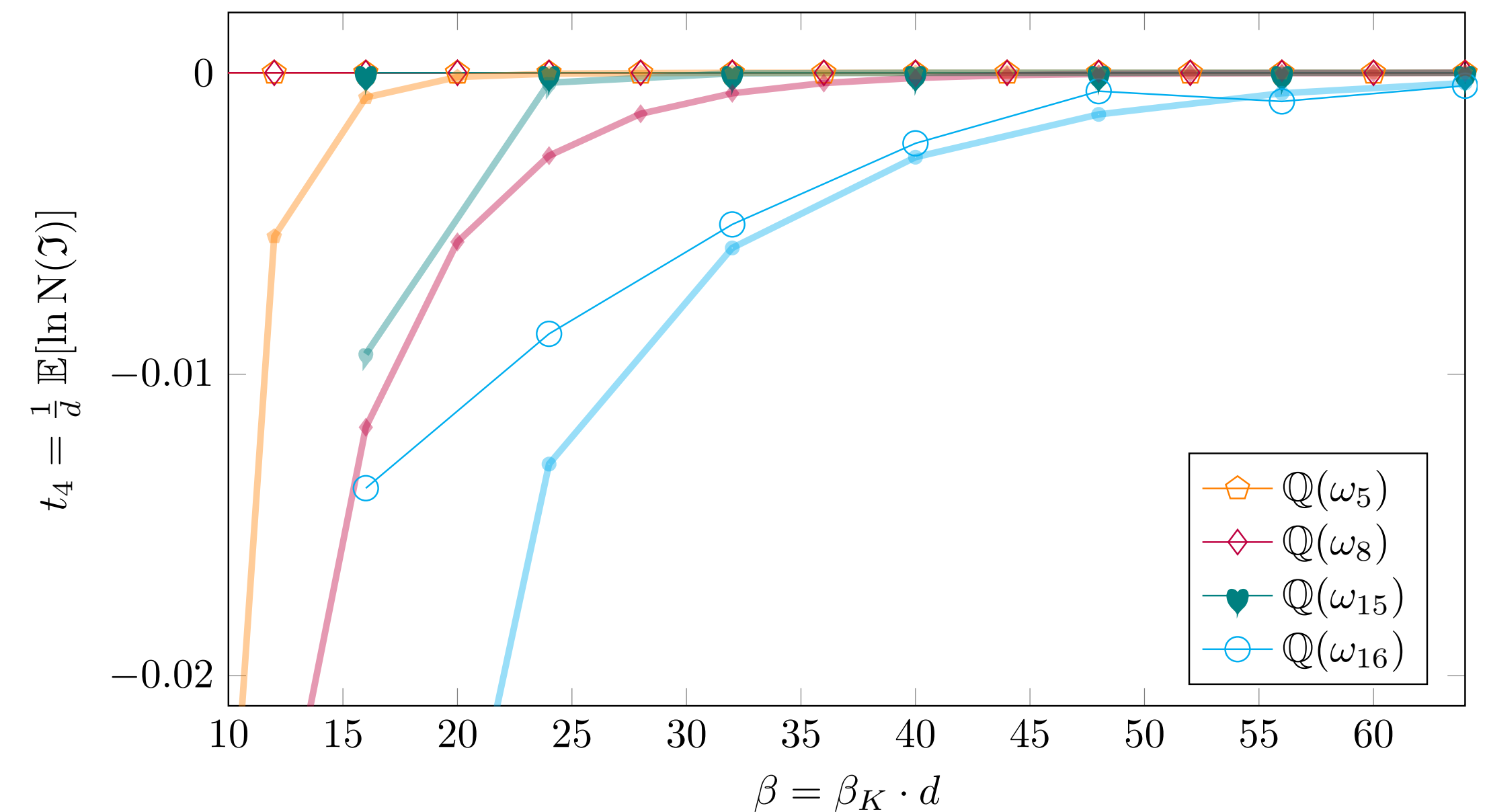
- t_4 measures how much “larger” I is
- $N(I)^{-1}$ is the **index** of I^{-1} as a subgroup of $\mathcal{O}_K$

We obtain a **prediction** $\tilde{t}_4(K, \beta_K)$ in terms of β_K and d

- Based on a (heuristic) density-based argument
 - Using the Dedekind zeta function of K

Our experiments suggest:

$$\tilde{t}_4(K, \beta_K) \leq t_4 \leq 0$$



thick lines: prediction $\tilde{t}_4(K, \beta_K)$

thin lines: experimental average (over 1000 samples)

Performance of module-BKZ

Heuristic prediction intervals:

For an $\text{mBKZ}_K^{\beta_K}$ -reduced basis:

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_K}) = -\frac{2}{\beta_K d - d} (t_1 + t_2 + t_3 + t_4)$$

where:

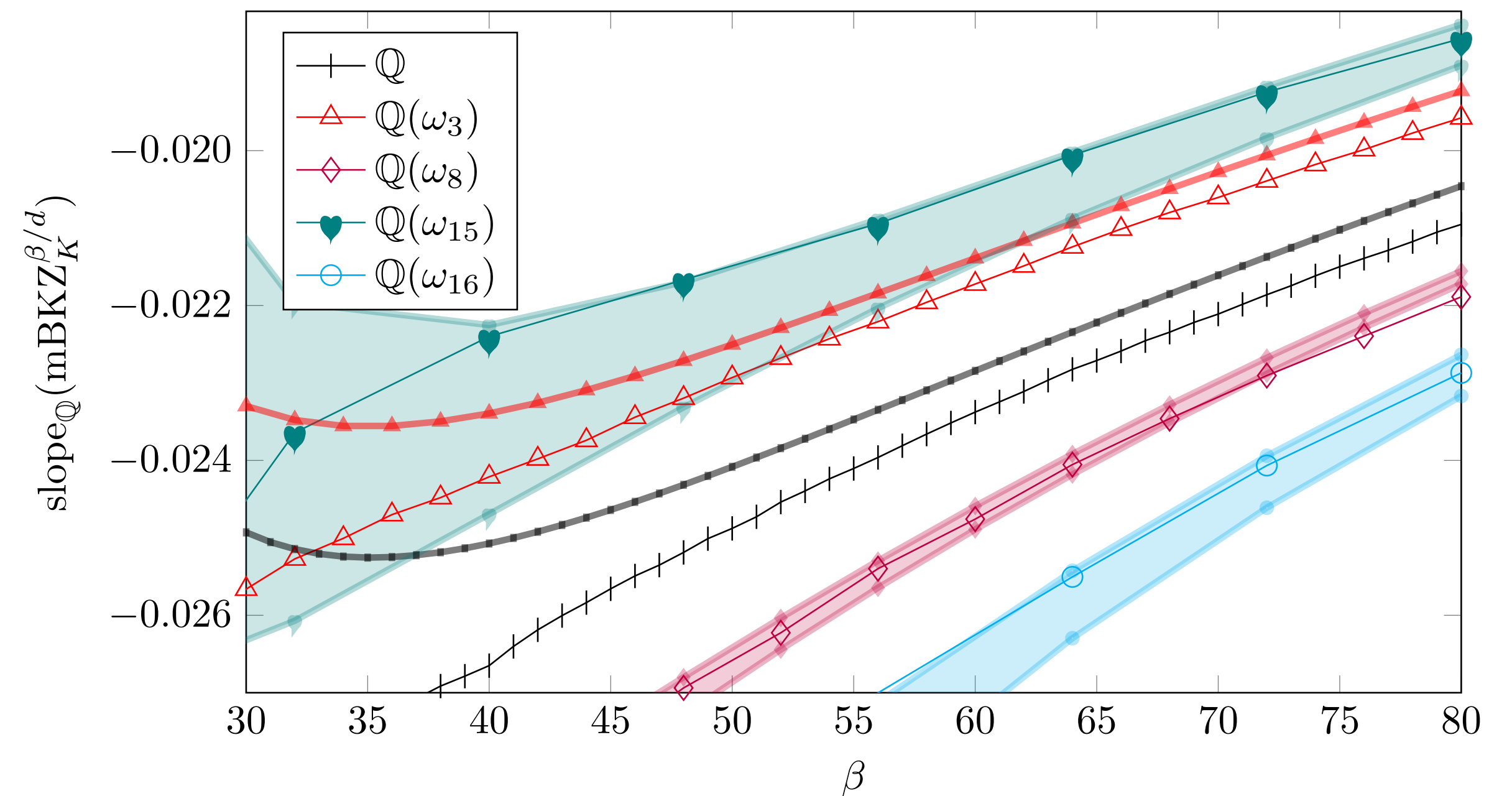
$$t_1 = \text{lgh}_{\mathbb{Q}}(\beta_K d) + \frac{1}{\beta_K d} \ln \frac{\mu_K}{2}$$

$$t_2 = \frac{1}{2d} \ln \frac{|\Delta_K|}{d^d}$$

$$\tilde{t}_3(K, \beta_K) \leq t_3 \leq 0$$

$$\tilde{t}_4(K, \beta_K) \leq t_4 \leq 0$$

Compared with experiments:



thick lines: prediction intervals

*thin lines: experimental average**

*over 5 random lattices of dimension $rd = 240$

Observation for $K = \mathbb{Q}(\omega_c)$:

Module-BKZ outperforms BKZ unless c is a power of two

Analysis of the blocksize gain

What can we say about the blocksize gain or loss?

Given β , we want to find the ‘**equivalent**’ blocksize β_{eq} such that:

$$\text{slope}_{\mathbb{Q}}(\text{mBKZ}_K^{\beta_{\text{eq}}/d}) = \text{slope}_{\mathbb{Q}}(\text{BKZ}^{\beta})$$

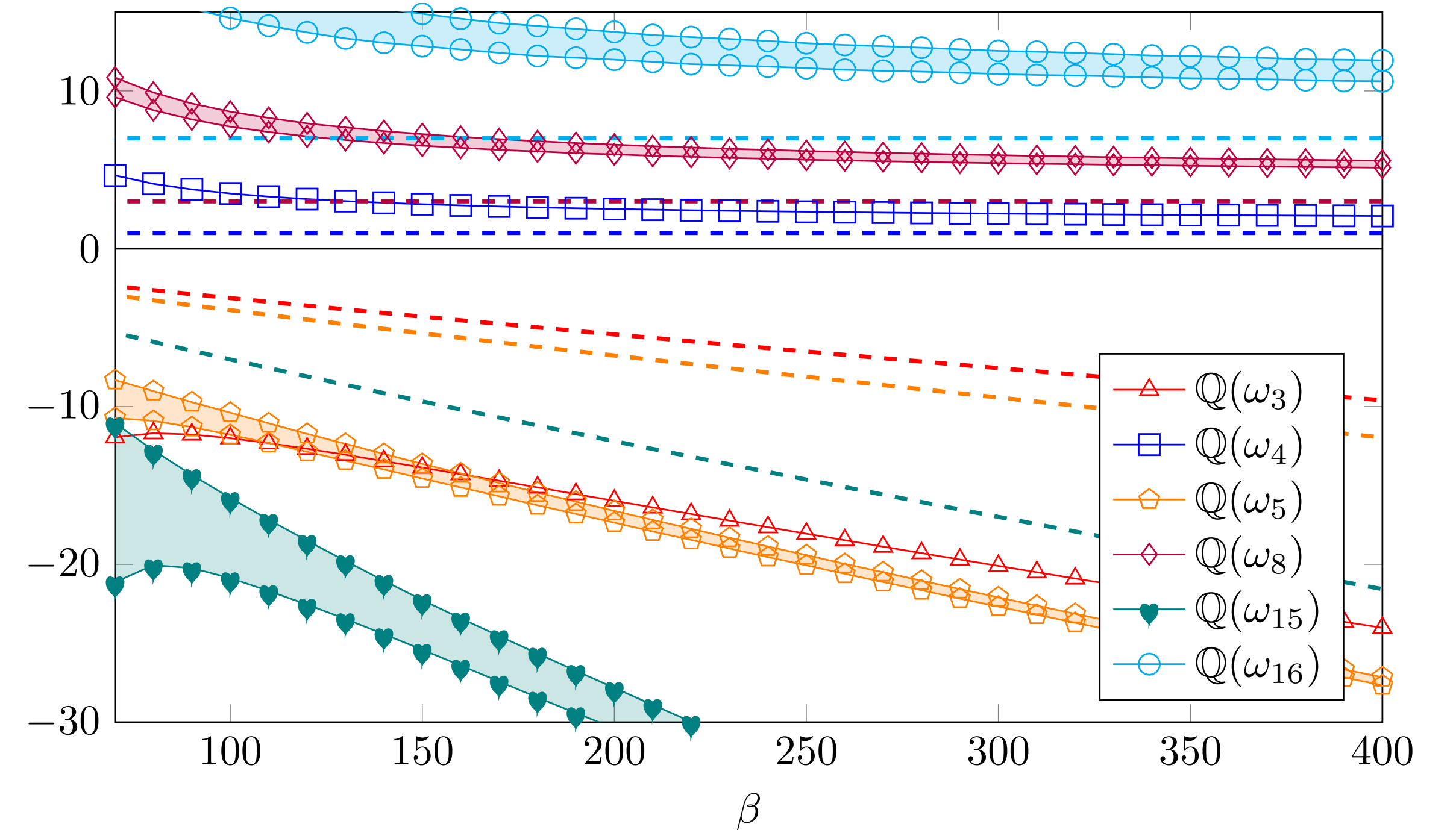
Asymptotic prediction: Main contributors are t_1 and t_2

If $|\Delta_K| = d^d$, then: $\beta_{\text{eq}} = \beta + d - 1 + o(1)$

▸ Module-BKZ requires a **larger** blocksize by at most a constant

If $|\Delta_K| < d^d$, then: $\beta_{\text{eq}} = \beta + \ln\left(\frac{|\Delta_K|}{d^d}\right) \frac{\beta}{d \ln \beta} + o\left(\frac{\beta}{\ln \beta}\right)$

▸ Module-BKZ requires a **smaller** blocksize, achieving a **sublinear gain**



lines with marks:
dashed lines:

prediction intervals
asymptotic prediction

Observation for $K = \mathbb{Q}(\omega_c)$: Asymptotic prediction underestimates the actual gain (or loss)

Impact

Consequences for cryptanalysis and open questions

Cryptanalytic impact

in practice

*Does module-BKZ perform **better** or **worse** than unstructured BKZ? By how much?*

Answer: It depends on the **number field** (key quantity: discriminant)

For a cyclotomic field $K = \mathbb{Q}(\omega_c)$:

- If $c > 2$ is a power of two: answer* = **worse** (constant loss)
- Otherwise: answer = **better** (slightly sublinear gain)

**Disclaimer: More work needed*

Consequence for module-lattice-based cryptography:

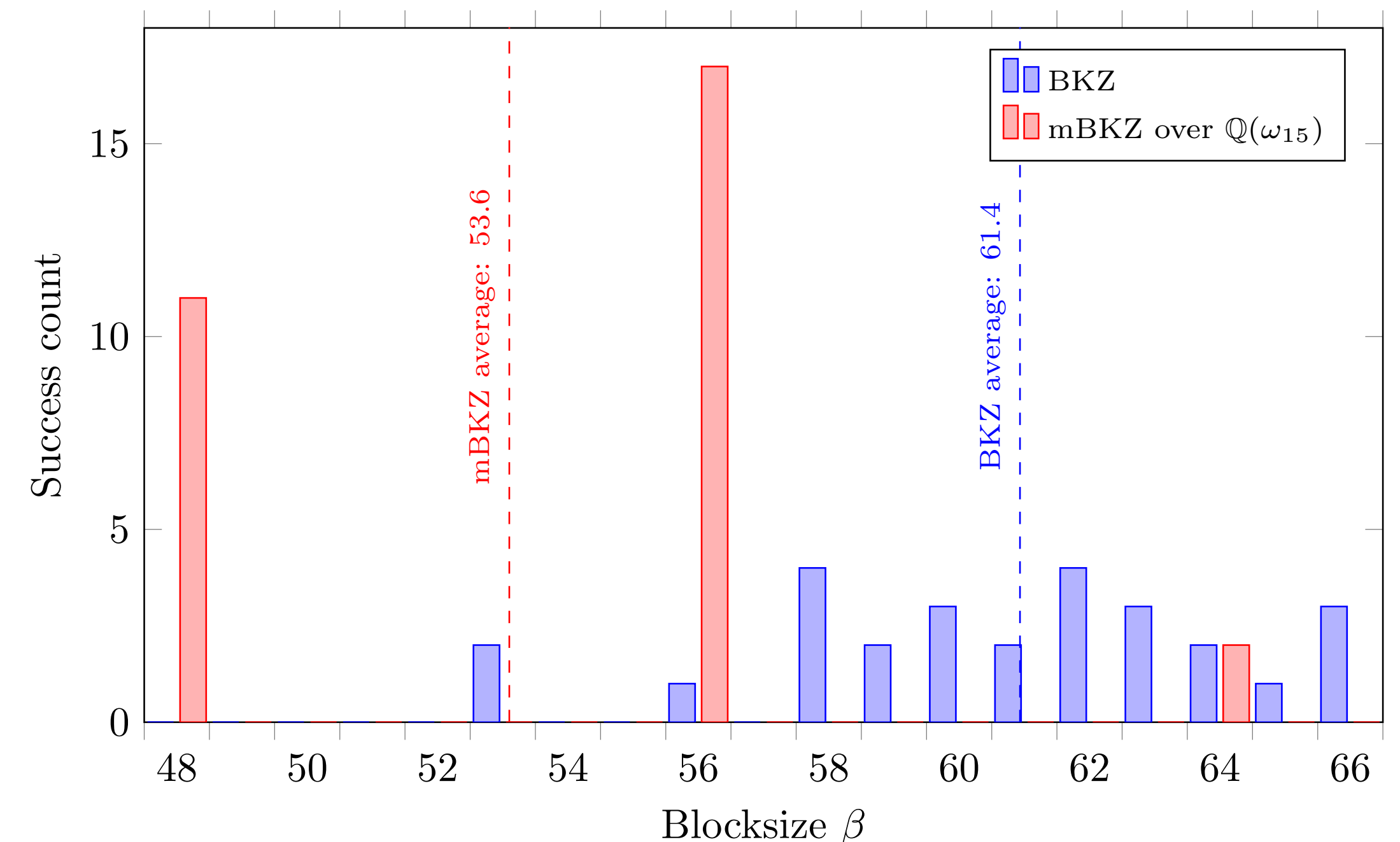
- For now, no speedup for NIST standards (Kyber, Dilithium, Falcon)
- Dimension gain for proposed schemes that work with $K = \mathbb{Q}(\omega_c)$ where c has a **small odd prime factor**
- Security estimates must account for the **number field choice** (including its subfields!)

Open questions

These questions would all benefit from a module-lattice reduction library :-)

1. Predicting **module-HKZ shape**, possibly giving a **tail-adapted refinement** of the module-lattice GSA
 - May also affect the cost of the SVP oracle by modifying the number of **dimensions for free**
2. Module-BKZ **profile simulation** to fine-tune attack and security estimates
3. **Advanced sieving**: SVP oracle may benefit from the cyclotomic structure
 - May change the “no” for power-of-two cyclotomic fields
4. Coefficient embedding (instead of canonical embedding)
5. SVP oracle versus **densest-ideal oracle**
 - Would this improve the slope?
 - How to efficiently implement such an oracle in practice?
6. Applications to solving module-lattice problems
 - Does the slope gain **translate into faster attacks**?

Module-LWE (30 experiments)
Secret and error dimension 136



Conclusion

Conclusion

Contributions:

- **Predictive framework** for module-lattice reduction, based on module-lattice analogs of the GSA and Gaussian Heuristic
- **Experiments** supporting the predictions, using an **implementation** of module-BKZ over several cyclotomic fields

Takeaways:

- Module-BKZ naturally suggests a **module-lattice GSA** for its K -profile, which allows to predict its **slope**
 - Module-BKZ is not universally better, as its performance depends on the number field (key driver: **discriminant**)
 - For power-of-two cyclotomic fields there is *no direct advantage* (in fact, a constant blocksize loss)
 - For other cyclotomic fields there is a *sublinear gain*
-  *Partial answer to Kyber's “Q8”*

Outlook:

- Many **open questions** remain
- Altogether this may help improve **security estimates** and guide future **scheme design**

Thank you!